

Literature-implied partial answer to a convex-transitive Banach–Mazur question

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2026-06-28

Status. This is a **literature-implied partial answer**, not a new proof packet.

Source question. Randrianantoanina’s *A note on Banach–Mazur problem* (arXiv:math/0110202) proves that a real Banach space of dimension at least three with an almost transitive isometry group and a 1-complemented hyperplane is Hilbert. In the introduction, after discussing the complex convex transitive reflection-vector theorem, the paper asks whether this theorem can be generalized to convex transitive spaces. In source lines 330–331:

“We do not know whether Theorem 1 can be generalized for convex transitive spaces.”

Supporting theorem. Talponen’s *Convex-transitive characterizations of Hilbert spaces* (arXiv:0705.2526) explicitly identifies the question as Randrianantoanina’s Question ($\mathbf{Q}_R$): whether every real convex-transitive Banach space containing a 1-codimensional 1-complemented subspace is Hilbert. The paper then proves affirmative subcases.

First, Theorem 3.2 says that if X is weakly locally uniformly rotund, has a big point, and for every $\varepsilon > 0$ has a 1-codimensional $(1 + \varepsilon)$ -complemented subspace, then X is Hilbert. A convex transitive space has every sphere point big, so this applies to the source projection hypothesis under the additional weak-LUR assumption.

Second, Theorem 5.4 says that if X is convex-transitive, admits a one-dimensional bicontractive projection, and has a weak-LUR point on the unit sphere, then X is Hilbert. This is a sharper weak-geometry subcase, but the bicontractive line projection is stronger than merely having a 1-complemented hyperplane.

Scope. The general convex-transitive variant of Randrianantoanina’s theorem remains open here. The supporting paper supplies substantial partial affirmative answers under weak-LUR or bicontractive projection hypotheses; it does not settle the unrestricted convex-transitive case.

Novelty check. The run registry had no existing packet for arXiv:0110202 or arXiv:0705.2526. It did contain a separate rotation-problem packet for the ℓ_p and subspace-of-quotient-of- ℓ_p classes, which is disjoint from this Flinn/bicontractive-projection subcase. Web and local searches for “1-complemented hyperplane convex transitive”, “Flinn element convex transitive”, and the source title found Talponen’s paper as the relevant later partial answer, with no full answer to the unrestricted question.

References

- [1] B. Randrianantoanina, *A note on Banach–Mazur problem*, Glasgow Math. J. 44 (2002), 159–165; arXiv:math/0110202.
- [2] J. Talponen, *Convex-transitive characterizations of Hilbert spaces*, arXiv:0705.2526.